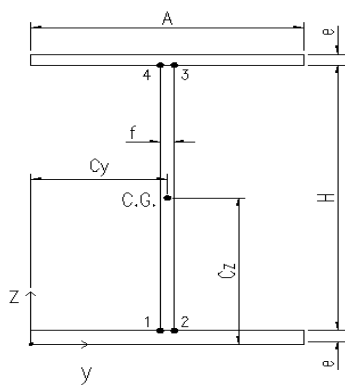


Test data

Section : I-Section



Geometry Parameters

A	100
H	210
e	20
f	12

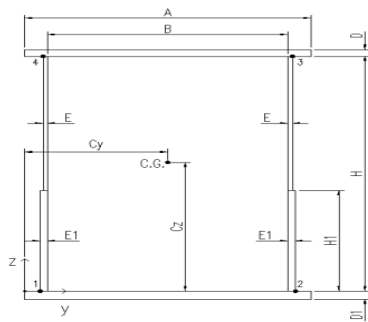
Properties

Area	6.5200E+03
Jz	3.3636E+06
Jy	6.2294E+07
Jx	6.5429E+05
Az	2.5200E+03
Ay	4.0000E+03
cy	50.0
cz	125.0

Stress points in the centroidal pri

No.	y
1	-6.0
2	6.0
3	6.0
4	-6.0

Section : Hollow Box



Geometry Parameters

A	1320
B	1250
H	2600
D	40
E	16
H1	600
D1	22
E1	16

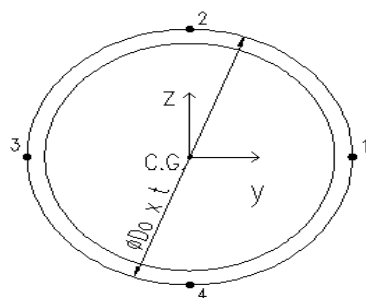
Properties

Area	1.6504E+05
Jz	4.5222E+10
Jy	1.8273E+11
Jx	1.0615E+11
Az	8.3200E+04
Ay	7.7500E+04
cy	660.0
cz	1491.6

Stress points in the centroidal pri

No.	y
1	-641.0

Section : Pipe



2	641.0
3	641.0
4	-641.0

Geometry Parameters

Do	1300
t	14

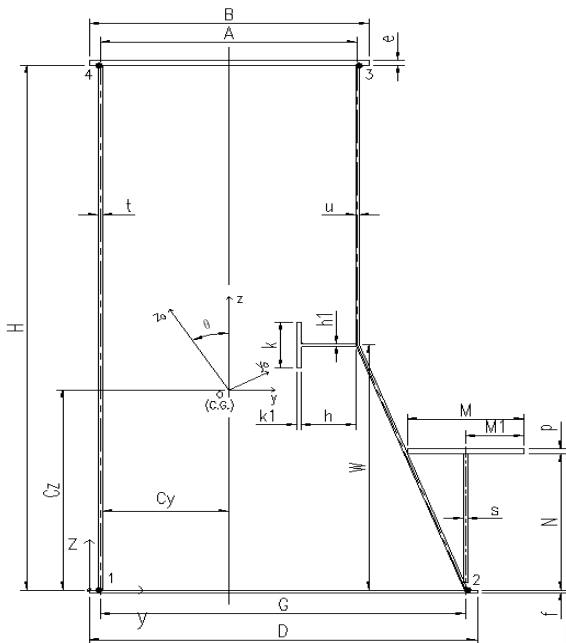
Properties

Area	5.6561E+04
Jz	1.1694E+10
Jy	1.1694E+10
Jx	2.3388E+10
Az	2.8283E+04
Ay	2.8283E+04
cy	0.0
cz	0.0

Stress points in the centroidal pri

No.	y
1	650.0
2	0.0
3	-650.0
4	0.0

Section : Crane Runway



Geometry Parameters

A	766
B	836
G	1090
D	1160
e	20
f	12
H	2000
W	934
M	350
N	518
p	20
s	12
t	10
u	10
M1	175
k	150
k1	12
h	138
h1	10

Properties

Area	8.7174E+04
Jz	1.4889E+10

Jy	4.9105E+10
Jx	2.1432E+10
Jyz	-8.0260E+09
θ	2.1933E-01
Jzo	1.3099E+10
Jyo	5.0894E+10
Az0	5.1147E+04
Ay0	5.4447E+04
cy	543.2
cz	941.2

Stress points in the centroidal pri

No.	yo
1	-735.0
2	328.9
3	447.8
4	-299.9

Output Cross-Section F

mm
mm
mm
mm

Area
Jy
Jz
Jx
cy
cz

mm2
mm4
mm4
mm4
mm2
mm2
mm
mm

Az
Ay
x_se
y_se
J

ncipal coordinate system

z
-105.0
-105.0
105.0
105.0

mm
mm
mm
mm
mm
mm
mm
mm

mm2
mm4
mm4
mm4
mm2
mm2
mm
mm

ncipal coordinate system

z
-1491.6

-1491.6
1108.4
1108.4

mm
mm

mm2
mm4
mm4
mm4
mm2
mm2
mm
mm

ncipal coordinate system

z
0.0
650.0
0.0
-650.0

mm
mm
mm
mm
mm
mm
mm
mm
mm
mm
mm
mm
mm
mm
mm
mm
mm
mm
mm
mm

mm2
mm4

mm4
mm4
mm4
rad

mm
mm
ncipal coordinate system

z0
-800.5
-1037.6
985.0
1151.6

Properties (Compare with Left Test Data)

SectionProperties

Section.calculate_geometric_properties
area (float None) – Cross-sectional area
ixx_c (float None) – Second moment of area about the centroidal x-axis
iyy_c (float None) – Second moment of area about the centroidal y-axis
ixy_c (float None) – Second moment of area about the centroidal xy-axis
cx (float None) – X coordinate of the elastic centroid
cy (float None) – Y coordinate of the elastic centroid
Section.calculate_warping_properties
a_sy (float None) – Shear area about the y-axis
a_sx (float None) – Shear area about the x-axis
x_se (float None) – x-coordinate of the shear centre (elasticity approach)
y_se (float None) – y-coordinate of the shear centre (elasticity approach)
j (float None) – Torsion constant

Geometric Properties

Below is a list of the section properties calculated by the method: `calculate_geometric_properties()`

- Cross-sectional area
- Cross-sectional perimeter
- Cross-sectional mass
- Area weighted material properties (composite only) $E_{eff}G_{eff}V_{eff}$
- Modulus weighted area (axial rigidity)
- First moments of area
- Second moments of area about the global axis
- Second moments of area about the centroidal axis
- Elastic centroid
- Centroidal section moduli
- Radii of gyration
- Principal axis properties
- Yield moments (composite only)

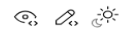
Warping Properties

Below is a list of the section properties calculated by the method: `calculate_warping_properties()`

- Torsion constant
- Shear centre
- Shear area
- Warping constant
- Monosymmetry constant

Class `SectionProperties`

SectionProperties



```
class sectionproperties.post.post.SectionProperties(area: float | None = None, perimeter: float | None = None, mass: float | None = None, ea: float | None = None, ga: float | None = None, nu_eff: float | None = None, e_eff: float | None = None, g_eff: float | None = None, qx: float | None = None, qy: float | None = None, ixx_g: float | None = None, iyy_g: float | None = None, ixy_g: float | None = None, cx: float | None = None, cy: float | None = None, ixx_c: float | None = None, iyy_c: float | None = None, ixy_c: float | None = None, zxx_plus: float | None = None, zxx_minus: float | None = None, zyy_plus: float | None = None, zyy_minus: float | None = None, rx_c: float | None = None, ry_c: float | None = None, i11_c: float | None = None, i22_c: float | None = None, phi: float | None = None, z11_plus: float | None = None, z11_minus: float | None = None, z22_plus: float | None = None, z22_minus: float | None = None, r11_c: float | None = None, r22_c: float | None = None, j: float | None = None, my_xx: float | None = None, my_yy: float | None = None, my_11: float | None = None, my_22: float | None = None, omega: ndarray[tuple[Any, ...], dtype[float64]] | None = None, psi_shear: ndarray[tuple[Any, ...], dtype[float64]] | None = None, phi_shear: ndarray[tuple[Any, ...], dtype[float64]] | None = None, delta_s: float | None = None, x_se: float | None = None, y_se: float | None = None, x11_se: float | None = None, y22_se: float | None = None, x_st: float | None = None, y_st: float | None = None, gamma: float | None = None, a_sx: float | None = None, a_sy: float | None = None, a_sxy: float | None = None, a_s11: float | None = None, a_s22: float | None = None, beta_x_plus: float | None = None, beta_x_minus: float | None = None, beta_y_plus: float | None = None, beta_y_minus: float | None = None, beta_11_plus: float | None = None, beta_11_minus: float | None = None, beta_22_plus: float | None = None, beta_22_minus: float | None = None, x_pc: float | None = None, y_pc: float | None = None, x11_pc: float | None = None, y22_pc: float | None = None, sxx: float | None = None, syy: float | None = None, sf_xx_plus: float | None = None, sf_xx_minus: float | None = None, sf_yy_plus: float | None = None, sf_yy_minus: float | None = None, s11: float | None = None, s22: float | None = None, sf_11_plus: float | None = None, sf_11_minus: float | None = None, sf_22_plus: float | None = None, sf_22_minus: float | None = None)
```

[\[source\]](#)

Bases: `object`

Class for storing section properties.

Stores calculated section properties. Also provides methods to calculate section properties externally.

stores calculated section properties. Also provides methods to calculate section properties entirely derived from other section properties.

